13-03-2026

Conditional Statement_50 Q & A

EASY LEVEL (1 - 20)

1. Write a program to check if a number is even or odd

In [1]:
```
x = int(input('enter a number:'))
r = x % 2

if r == 0:
    print('Even number')
else:
    print('Odd number')
```

```
Even number
```

2. Check if a person is eligible to vote (age 18 or above).

In [2]:
```
age = int(input('enter your age:'))
a = age

if a>=18:
    print('You are eligible to vote')
else:
    print('You are not eligible to vote')
```

```
You are eligible to vote
```

3. Determine if a given year is a leap year or not

In [3]:
```
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
 print("Leap Year")
else:
 print("Not a Leap Year")
```

```
Not a Leap Year
```

4. Check if a given number is positive, negative or zero

In [4]:
```
num = int(input('enter a number:'))
if num < 0:
    print('given number is -ve')
elif num == 0:
    print('given number is zero')
else:
    print('given number is +ve')
```

```
given number is +ve
```

5. Write a program to find the greatest of two numbers

In [5]:
```
n1 = int(input('enter first number:'))
n2 = int(input('enter second number:'))
if n1>n2:
    print('first number is greater than second number')
else:
    print('second number is greater than first number')
```

```
second number is greater than first number
```

6. Determine if a number is a multiple of 5

In [6]:
```
x = int(input('enter a number:'))
if x % 5 == 0:
    print('given number is multiple of 5')
else:
    print('given number is not multiple of 5')
```

```
given number is not multiple of 5
```

7. Check if a character is a vowel or consonant

In [7]:
```
c = input('enter a letter:')
if c in 'aeiouAEIOU':
    print('given letter is a vowel')
else:
    print('given letter is a consonant')
```

```
given letter is a vowel
```

8. Determine if a person is eligible for a senior citizen discount (age 60+)

In [8]:
```
age = int(input('enter your age:'))
a = age

if a>=60:
    print('You are eligible for a senior citizen discount')
else:
    print('You are not eligible for a senior citizen discount')
```

```
You are eligible for a senior citizen discount
```

9. Write a program to check if a number is a single-digit number.

In [9]:
```
num = int(input('Enter a number:'))
if 0 <= num < 10:
    print('Single-digit number')
else:
    print('Not a single-digit number')
```

```
Not a single-digit number
```

10. Print "Good Morning if the time is before 12 PM, otherwise print "Good Afternoon".

In [10]:
```
hour = int(input("Enter hour (24-hour format): "))
if hour < 12:
    print("Good Morning")
else:
    print("Good Afternoon")
```

```
Good Afternoon
```

11. Check if a string is empty or not.

In [11]:
```
s = input("Enter a string: ")
if not s:
    print("String is empty")
else:
    print("String is not empty")
```

```
String is empty
```

12. Verify if a number is a perfect square.

In [12]:
```
import math
num = int(input("Enter a number: "))
if math.isqrt(num) ** 2 == num:
    print("Perfect Square")
```

```
else:
    print("Not a perfect square")
```

```
Not a perfect square
```

13. Determine if a number is between 1 and 100.

In [13]:
```
num = int(input("Enter a number: "))
if 1 <= num <= 100:
    print("Within range")
else:
    print("Out of range")
```

```
Within range
```

14. Print "Weekend" if the day is Saturday or Sunday; otherwise, print "Weekday".

In [14]:
```
day = input("Enter a day: ").lower()
if day in ["saturday", "sunday"]:
    print("Weekend")
else:
    print("Weekday")
```

```
Weekend
```

15. Find if a given number is exactly divisible by both 3 and 7.

In [15]:
```
num = int(input("Enter a number: "))
if num % 3 == 0 and num % 7 == 0:
    print("Divisible by 3 and 7")
else:
    print("Not divisible by both")
```

```
Not divisible by both
```

16. Check if the sum of two numbers is greater than 100.

In [16]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
if (a + b) > 100:
    print("Sum is greater than 100")
elif (a + b) == 100:
    print("Sum is equals to 100")
else:
    print("Sum is lessthan 100")
```

```
Sum is greater than 100
```

17. Write a program to find the minimum of two numbers.

In [17]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
if a < b:
    print("Minimum:", a)
else:
    print("Minimum:", b)
```

```
Minimum: 15
```

18. Check if a number is divisible by 2 but not by 3.

In [18]:
```
num = int(input("Enter a number: "))
if num % 2 == 0 and num % 3 != 0:
    print("Divisible by 2 but not by 3")
else:
    print("Does not meet criteria")
```

```
Divisible by 2 but not by 3
```

19. Determine if a given alphabet is uppercase or lowercase.

In [19]:
```
char = input("Enter an alphabet: ")
if char.isupper():
    print("Uppercase")
else:
    print("Lowercase")
```

```
Uppercase
```

20. Check if a triangle is valid given three side lengths (sum of any two sides must be greater than the third).

In [20]:
```
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
if a + b > c and a + c > b and b + c > a:
    print("Valid Triangle")
else:
    print("Invalid Triangle")
```

```
Invalid Triangle
```

MEDIUM LEVEL (21 - 40)

21. Find the largest of three numbers.

In [21]:
```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
    print("Largest:", a)
elif b >= a and b >= c:
    print("Largest:", b)
else:
    print("Largest:", c)
```

```
Largest: 85
```

22. Determine if a number is a prime number.

In [22]:
```
num = int(input("Enter a number: "))
if num > 1:
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            print("Not a prime number")
            break
        else:
            print("Prime number")
    else:
        print("Not a prime number")
```

```
Not a prime number
```

23. Check if a person is eligible for a driving license (age 18+, passed the driving test).

In [23]:
```
age = int(input("Enter age: "))
passed_test = input("Did you pass the driving test? (yes/no): ").lower()
if age >= 18 and passed_test == "yes":
    print("Eligible for a driving license")
else:
    print("Not eligible")
```

```
Eligible for a driving license
```

24. Write a program to determine whether a triangle is equilateral, isosceles, or scalene.

In [24]:
```
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
if a == b == c:
    print("Equilateral Triangle")
```

```
elif a == b or b == c or a == c:
    print("Isosceles Triangle")
else:
    print("Scalene Triangle")
```

```
Scalene Triangle
```

25. Determine if a student passes or fails based on a passing mark of 40.

In [25]:
```
marks = int(input("Enter marks:"))
if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

```
Pass
```

26. Check if a number is a palindrome (same forward and backward).

In [26]:
```
num = input("Enter a number: ")
if num == num[::-1]:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

```
Palindrome
```

27. Calculate the electricity bill based on consumption: ₹5 per unit for the first 100 units, ₹10 per unit for the next 200, ₹15 per unit for anything above 300 units.

In [27]:
```
units = int(input("Enter electricity units consumed: "))
if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = (100 * 5) + ((units - 100) * 10)
else:
    bill = (100 * 5) + (200 * 10) + ((units - 200) * 15)
print("Total Bill: ₹", bill)
```

```
Total Bill: ₹ 3610
```

28. Find the grade of a student based on marks (90+ A, 80-89 B, etc.).

In [28]:
```
marks = int(input("Enter marks: "))
if marks >= 90:
    print("Grade: A")
elif marks >= 80:
    print("Grade: B")
elif marks >= 70:
    print("Grade: C")
elif marks >= 60:
    print("Grade: D")
elif marks >= 40:
    print("Grade: E")
else:
    print("Grade: F (Fail)")
```

```
Grade: A
```

29. Determine if a given date is valid (considering month length and leap year for February).

In [29]:
```
import calendar
day = int(input("Enter day: "))
month = int(input("Enter month: "))
year = int(input("Enter year: "))
if 1 <= month <= 12 and 1 <= day <= calendar.monthrange(year, month)[1]:
    print("Valid date")
else:
    print("Invalid date")
```

```
Valid date
```

30. Check if a given time is AM or PM.

In [30]:
```
hour = int(input("Enter hour (24-hour format): "))
if hour < 12:
    print("AM")
else:
    print("PM")
```

```
AM
```

31. Check if a number is an Armstrong number (sum of its digits raised to the power of the number of digits equals the number).

In [31]:
```
num = input("Enter a number:")
power = len(num)
if sum(int(digit) ** power for digit in num) == int(num):
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

```
Not an Armstrong Number
```

32. Determine the type of quadrilateral based on given angles and sides.

In [32]:
```
a = int(input('Enter first side:'))
b = int(input('Enter second side:'))
c = int(input('Enter third side:'))
d = int(input('Enter fourth side:'))
if a == b == c == d:
    print('Square')
elif a == c & b == d:
    print('Rectangle')
else:
    print('Other Quadrilateral')
```

```
Square
```

33. Implement a basic calculator that takes two numbers and an operation (+, -, *, /).

In [33]:
```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
op = input("Enter operation (+, -, *, /): ")
if op == "+":
    print("Result:", a + b)
elif op == "-":
    print("Result:", a - b)
elif op == "*":
    print("Result:", a * b)
elif op == "/":
    print("Result:", a / b)
else:
    print("Invalid operation")
```

```
Result: 70.65
```

34. Check if a bank account balance is sufficient for withdrawal.

In [34]:
```
balance = float(input("Enter account balance: "))
withdraw = float(input("Enter withdrawal amount: "))
if balance >= withdraw:
    print("Withdrawal successful")
else:
    print("Insufficient funds")
```

```
Withdrawal successful
```

35. Implement a simple temperature converter (Celsius to Fahrenheit and vice versa).

In [35]:
```
temp = float(input("Enter temperature: "))
unit = input("Enter unit (C/F): ").upper()
if unit == "C":
    print("Fahrenheit:", (temp * 9/5) + 32)
elif unit == "F":
    print("Celsius:", (temp - 32) * 5/9)
else:
    print("Invalid unit")
```

```
Fahrenheit: 78.44
```

36. Write a program to check if a number lies within a specific range (e.g., between 50 and 100).

In [36]:
```
num = int(input("Enter a number: "))
if num in range(50, 101):
    print("Within range")
else:
    print("Out of range")
```

```
Within range
```

37. Determine if a given year is a century year (ends in 00).

In [37]:
```
year = int(input('Enter a year:'))
if year % 100 == 0:
    print('Century Year')
else:
    print('Not a Century Year')
```

```
Not a Century Year
```

38. Check if a given integer is a power of 2.

In [38]:
```
num = int(input('Enter a number:'))
if num > 0 & (num & (num - 1)) == 0:
    print('Power of 2')
else:
    print('Not a Power of 2')
```

```
Power of 2
```

39. Determine whether a given month has 31, 30, or 28 days.

In [39]:
```
month = int(input("Enter month (1-12): "))
year = int(input("Enter year: "))
days = [31, 28 + (1 if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0) else 0),
        31, 30, 31, 30, 31, 31, 30, 31, 30, 31]
print("Days:", days[month - 1])
```

```
Days: 31
```

40. Validate a password (must be at least 8 characters long and contain both numbers and letters).

In [40]:
```
import re
password = input("Enter password: ")
if len(password) >= 8 and re.search(r"[A-Za-z]", password) and re.search(r"\d", password):
    print("Valid Password")
else:
    print("Invalid Password")
```

```
Valid Password
```

HARD LEVEL (41 - 50)

41. Implement a ticket pricing system where children under 5 enter for free, seniors (60+) get a 50% discount, and others pay ₹100.

In [41]:
```
age = int(input("Enter age: "))
if age < 5:
    print("Ticket Price: Free")
elif age >= 60:
    print("Ticket Price: ₹50")
else:
    print("Ticket Price: ₹100")
```

```
Ticket Price: ₹100
```

42. Check if three numbers can form a Pythagorean triplet.

In [42]:
```
a, b, c = sorted(map(int, input("Enter three numbers: ").split()))
if a**2 + b**2 == c**2:
    print("Pythagorean Triplet")
else:
    print("Not a Pythagorean Triplet")
```

```
Pythagorean Triplet
```

43. Write a program that converts a Roman numeral to an integer.

In [43]:
```
def roman_to_int(s):
    roman = {'I': 1, 'V': 5, 'X': 10, 'L': 50, 'C': 100, 'D': 500, 'M': 1000}
    total = 0
    for i in range(len(s)):
        if i > 0 and roman[s[i]] > roman[s[i - 1]]:
            total += roman[s[i]] - 2 * roman[s[i - 1]]
        else:
            total += roman[s[i]]
    return total
num = input("Enter Roman numeral: ").upper()
print("Integer:", roman_to_int(num))
```

```
Integer: 1500
```

44. Determine the zodiac sign based on a given birth date.

In [44]:
```
month = int(input("Enter birth month (1-12): "))
day = int(input("Enter birth day: "))
zodiac = [("Capricorn", 20), ("Aquarius", 19), ("Pisces", 20), ("Aries", 20),
("Taurus", 21), ("Gemini", 21), ("Cancer", 22), ("Leo", 22),
("Virgo", 22), ("Libra", 23), ("Scorpio", 23), ("Sagittarius", 22), ("Capricorn", 31)]
if day <= zodiac[month - 1][1]:
    sign = zodiac[month - 1][0]
else:
    sign = zodiac[month][0]
print("Zodiac Sign:", sign)
```

```
Zodiac Sign: Taurus
```

45. Check if a number is a Harshad number (sum of digits divides the number).

In [45]:
```
num = int(input("Enter a number: "))
sum_digits = sum(int(digit) for digit in str(num))
if num % sum_digits == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")
```

```
Not a Harshad Number
```

46. Validate an email format (must contain '@' and end with a valid domain like '.com' or '.org').

In [46]:
```
import re
email = input("Enter email: ")
if re.match(r"^[\w\.-]+@[\w\.-]+\.(com|org|net|edu)$", email):
    print("Valid Email")
else:
    print("Invalid Email")
```

```
Valid Email
```

47. Check if a given chess move is valid (e.g., a knight moves in an "L" shape).

In [47]:
```
x1, y1 = map(int, input('Enter current position (x, y):').split())
x2, y2 = map(int, input('Enter current position (x, y):').split())
if (abs(x1 - x2), abs(y1 - y2)) in [(2, 1), (1, 2)]:
    print('Valid Knight Move')

else:
    print('Invalid Move')
```

```
Invalid Move
```

48. Implement a loan eligibility checker based on income, credit score, and employment status.

In [48]:
```
income = int(input('Enter monthly income:'))
credit_score = int(input('Enter credit score:'))
employed = input('Are you employed? (yes/no):').lower()
if income >= 25000 and credit_score >= 700 and employed == 'yes':
    print('Loan Approved')
else:
    print('Loan Denied')
```

```
Loan Approved
```

49. Implement a rock-paper-scissors game using conditional statements.

In [49]:
```
import random
choices = ['rock', 'paper', 'scissors']
user = input('Enter rock, paper or scissors:').lower()
computer = random.choice(choices)
print('Computer chose:', computer)
if user == computer:
    print("It's a tie!")
elif (user == 'rock' and computer == 'scissors') or\
(user == 'scissors' and computer == 'paper') or\
(user == 'paper' and computer == 'rock'):
    print('You WIN!')
else:
    print('You LOSE!')
```

```
Computer chose: scissors
You WIN!
```

50. Write a program to find the day of the week for a given date (without using built-in functions).

In [50]:
```
def day_of_week(d, m, y):
    if m < 3:
        m += 12
        y -= 1
    k = y % 100
    j = y // 100
    day = (d + (13 * (m + 1)) // 5 + k + (k // 4) + (j // 4) - 2 * j) % 7
    days = ["Saturday", "Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday"]
    return days[day]
day, month, year = map(int, input("Enter date (DD MM YYYY): ").split())
print("Day of the Week:", day_of_week(day, month, year))
```

```
Day of the Week: Wednesday
```